

Лабораторная работа № 3

Настройка DHCP-сервера

Павличенко Родион Андреевич

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1. Информация

1.1 Докладчик

- Павличенко Родион Андреевич
- Студент
- Группа НПИбд-02-24
- Российский университет дружбы народов
им. Патриса Лумумбы
- 1132246838@rudn.ru

2. Цель и задание

2.1 Цель работы

Приобрести практические навыки по установке и конфигурированию DHCP-сервера.

2.2 Задание

- установить и настроить Kea DHCP на машине `server`;
- проверить выдачу сетевых параметров машине `client`;
- настроить динамическое обновление DNS-зон;
- проверить работу DHCP и DDNS;
- автоматизировать настройку средствами Vagrant.

3. Настройка DHCP-сервера

3.1 Запуск серверной машины

Запущена виртуальная машина
server.

```
C:\work\rapavlichenko>vagrant up server
Bringing machine 'server' up with 'virtualbox' provider...
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
```

Выполнено подключение по SSH.

```
C:\work\rapavlichenko>vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please enter
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Tue Sep 15 21:27:56 2026 from 10.0.2.2
vagrant@server:~$
```

3.2 Получение административных прав

Получены полномочия суперпользователя для настройки системных служб.

```
vagrant@server:~$ sudo -i  
root@server:~#
```

3.3 Установка Kea DHCP

Установлен пакет DHCP-сервера Kea.

```
root@server:~# dnf -y install kea
Extra Packages for Enterprise Linux 10 - x86_64          34 kB/s | 29 kB
00:00
Extra Packages for Enterprise Linux 10 - x86_64          1.7 MB/s | 6.7 MB
00:03
Rocky Linux 10 - BaseOS                                   2.8 kB/s | 4.3 kB
00:01
Rocky Linux 10 - BaseOS                                   3.6 MB/s | 28 MB
00:07
Rocky Linux 10 - AppStream                                5.3 kB/s | 4.4 kB    00:00
Rocky Linux 10 - AppStream                               953 kB/s | 2.7 MB    00:02
Rocky Linux 10 - CRB                                     2.1 kB/s | 4.3 kB    00:02
Rocky Linux 10 - CRB                                     137 kB/s | 546 kB    00:03
Rocky Linux 10 - Extras                                   2.2 kB/s | 3.1 kB    00:01
Package kea-3.0.3-1.el10_2.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

3.4 Подготовка конфигурации

Создана резервная копия
исходной конфигурации.

```
Complete!  
root@server:~# cp /etc/kea/kea-dhcp4.conf "/etc/kea/kea-dhcp4.conf_$(date -I)"  
root@server:~#
```

Настроены подсеть, пул адресов,
DNS и параметры аренды.

```
GNU nano 8.1 /etc/kea  
{  
  "Dhcp4": {  
    "interfaces-config": {  
      "interfaces": [ "eth1" ]  
    },  
    "lease-database": {  
      "type": "memfile",  
      "persist": true,  
      "name": "/var/lib/kea/kea-leases4.csv"  
    },  
    "valid-lifetime": 4000,  
    "renew-timer": 1000,  
    "rebind-timer": 2000,  
    "option-data": [  
      {  
        "code": 15,  
        "data": "rapavlichenko.net"  
      },  
      {  
        "name": "domain-search",  
        "data": "rapavlichenko.net"  
      },  
      {  
        "name": "domain-name-servers",  
        "data": "192.168.1.1"  
      }  
    ],  
    "subnet4": [  
      {  
        "id": 1,  
        "subnet": "192.168.1.0/24",  
        "pools": [  
          {  
            "pool": "192.168.1.30 - 192.168.1.199"  
          }  
        ],  
        "option-data": [  
          {  
            "name": "routers",
```

3.5 Интерфейс и проверка конфигурации

Сервер привязан к интерфейсу eth1.

```
{
  "Dhcp4": {
    "interfaces-config": {
      "interfaces": [ "eth1" ]
    }
  }
}
```

Файл проверен средствами Кеа.

```
root@dhcp4:~# kea-dhcpd -t /etc/kea/kea-dhcpd.conf
2026-09-17 18:00:57.947 WARN [kea-dhcpd.dhcpd/13629.140578566400128] DHCPD_DISABLE_QUEUE_CONTROL disabling dhcp queue control when multi-threading is enabled
2026-09-17 18:00:57.954 WARN [kea-dhcpd.dhcpd/13629.140578566400128] DHCPD_RESERVATIONS_LOOKUP_FIRST_ENABLED Multi-threading is enabled and host reservations lookup is always performed first.
2026-09-17 18:00:57.961 INFO [kea-dhcpd.dhcpd/13629.140578566400128] DHCPD_CFGMGR_NEW_SUBNET4 a new subnet has been added to configuration: 192.168.1.0/24 with params: t1=1000, t2=2000, valid-lifetime=4000
2026-09-17 18:00:57.962 INFO [kea-dhcpd.dhcpd/13629.140578566400128] DHCPD_CFGMGR_SOCKET_TYPE_SELECT using socket type raw
2026-09-17 18:00:57.966 INFO [kea-dhcpd.dhcpd/13629.140578566400128] DHCPD_CFGMGR_ADD_IFACE listening on interface eth1
2026-09-17 18:00:57.970 INFO [kea-dhcpd.dhcpd/13629.140578566400128] DHCPD_CFGMGR_SOCKET_TYPE_DEFAULT "dhcp-socket-type" not specified, using default socket type raw
2026-09-17 18:00:57.970 INFO [kea-dhcpd.dhcpd/13629.140578566400128] DHCPD_LEASE_MGR_BACKENDS_REGISTERED the following lease backend types are available: memfile
2026-09-17 18:00:57.970 INFO [kea-dhcpd.hosts/13629.140578566400128] HOSTS_BACKENDS_REGISTERED the following host backend types are available:
2026-09-17 18:00:57.971 INFO [kea-dhcpd.dhcpd/13629.140578566400128] DHCPD_FORENSIC_BACKENDS_REGISTERED the following forensic backend types are available:
2026-09-17 18:00:57.971 INFO [kea-dhcpd.database/13629.140578566400128] CONFIG_BACKENDS_REGISTERED the following config backend types are available:
```

3.6 Автоматический запуск Kea

Служба kea-dhcp4 включена в автозагрузку.

```
root@server:~# systemctl --system daemon-reload  
root@server:~# systemctl enable kea-dhcp4.service  
root@server:~# |
```

4. Регистрация DHCP в DNS

4.1 Записи прямой и обратной зон

Добавлена А-запись dhcp.

```
GNU nano 8.1
$TTL 1D
@      IN SOA  @ server.rapavlichenko.net. (
                                2026091500 ; serial
                                1D          ; refresh
                                1H          ; retry
                                1W          ; expire
                                3H )        ; minimum

      NS      @
      A       192.168.1.1
$ORIGIN rapavlichenko.net.
server A       192.168.1.1
ns     A       192.168.1.1
dhcp   A       192.168.1.1
```

Добавлена PTR-запись
ДНСР-сервера.

```
GNU nano 8.1
$TTL 1D
@      IN SOA  @ server.rapavlichenko.net. (
                                2026091500 ; serial
                                1D          ; refresh
                                1H          ; retry
                                1W          ; expire
                                3H )        ; minimum

      NS      @
      A       192.168.1.1
      PTR     server.rapavlichenko.net.
$ORIGIN 1.168.192.in-addr.arpa.
1      PTR     server.rapavlichenko.net.
1      PTR     ns.rapavlichenko.net.
1      PTR     dhcp.rapavlichenko.net.|
```

4.2 Проверка имени DHCP-сервера

Имя `dhcp.rapavlichenko.net` разрешается в адрес `192.168.1.1`.

```
root@server:~# systemctl restart named
root@server:~# ping -c 4 dhcp.rapavlichenko.net
PING dhcp.rapavlichenko.net (192.168.1.1) 56(84) bytes of data.
64 bytes from server.rapavlichenko.net (192.168.1.1): icmp_seq=1 ttl=64 time=0.073 ms
64 bytes from server.rapavlichenko.net (192.168.1.1): icmp_seq=2 ttl=64 time=0.083 ms
64 bytes from server.rapavlichenko.net (192.168.1.1): icmp_seq=3 ttl=64 time=0.120 ms
64 bytes from server.rapavlichenko.net (192.168.1.1): icmp_seq=4 ttl=64 time=0.113 ms

--- dhcp.rapavlichenko.net ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3075ms
rtt min/avg/max/mdev = 0.073/0.097/0.120/0.019 ms
root@server:~# |
```

4.3 Межсетевой экран и SELinux

В firewalld разрешена служба DHCP.

```
root@server:~# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps mountd nfs ntp pop3 popls rpc-bind samba smtp smtp-submission ssh ssh-custom
root@server:~# firewall-cmd --get-services
firewalld
root@server:~# firewall-cmd --add-service=dhcp
success
root@server:~# firewall-cmd --add-service=dhcp --permanent
success
root@server:~# firewall-cmd --add-service=dhcp
success
```

Восстановлены контексты SELinux.

```
root@server:~# restorecon -vR /etc
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:obj
:s0
root@server:~# restorecon -vR /var/named
root@server:~# restorecon -vR /var/lib/kea/
root@server:~#
```

4.4 Запуск и наблюдение за службой

Запущен просмотр системного журнала.

```
C:\work\rapavlichenko>vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please enter
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Thu Sep 17 17:47:48 2026 from 10.0.2.2
vagrant@server:~$ sudo -i
root@server:~# tail -f /var/log/messages
Sep 17 18:11:34 server systemd-logind[1082]: New session 8 of user vagrant.
Sep 17 18:11:34 server systemd[1]: Started session-8.scope - Session 8 of User vagrant.
Sep 17 18:11:35 server systemd[1]: Starting systemd-hostnamed.service - Hostname Service...
Sep 17 18:11:35 server systemd[1]: Started systemd-hostnamed.service - Hostname Service.
Sep 17 18:11:40 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.
Sep 17 18:11:40 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter is at 172.
Sep 17 18:11:40 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.
Sep 17 18:11:50 server systemd[1]: serial-getty@ttyS0.service: Deactivated successfully.
Sep 17 18:11:50 server systemd[1]: serial-getty@ttyS0.service: Scheduled restart job, restart counter is at 173.
Sep 17 18:11:50 server systemd[1]: Started serial-getty@ttyS0.service - Serial Getty on ttyS0.
```

Запущена служба kea-dhcp4.

```
root@server:~# systemctl start kea-dhcp4.service
root@server:~#
```

5. Настройка клиента

5.1 Маршрутизация клиентской машины

Сценарий задаёт шлюз 192.168.1.1 на интерфейсе eth1 и отключает маршрут по умолчанию через eth0.

```
C: > work > rapavlichenko > provision > client > $ 01-routing.sh
1  #!/bin/bash
2  echo "Provisioning script $0"
3
4  nmcli connection modify "eth1" ipv4.gateway "192.168.1.1"
5  nmcli connection up "eth1"
6  nmcli connection modify eth0 ipv4.never-default true
7  nmcli connection modify eth0 ipv6.never-default true
8  nmcli connection down eth0
9  nmcli connection up eth0
```

5.2 Подключение сценария в Vagrant

В Vagrantfile добавлен provisioner client routing.

```
client.vm.provision "client routing",  
  type: "shell",  
  preserve_order: true,  
  run: "always",  
  path: "provision/client/01-routing.sh"  
end  
end
```

5.3 Запуск клиента

Клиентская машина запущена с повторным применением настроек.

```
C:\work\rapavlichenko>vagrant up client --provision
Bringing machine 'client' up with 'virtualbox' provider...
==> client: Clearing any previously set forwarded ports...
==> client: Fixed port collision for 22 => 2222. Now on port 2200.
==> client: Clearing any previously set network interfaces...
==> client: Preparing network interfaces based on configuration...
    client: Adapter 1: nat
    client: Adapter 2: intnet
==> client: Forwarding ports...
    client: 22 (guest) => 2200 (host) (adapter 1)
```

5.4 Выдача адреса клиенту

Журнал Кеа показывает запрос клиента, выделение адреса 192.168.1.30 и отправку DHCPACK.

```
ery: [hwtype=1 08:00:27:3b:eb:45], cid=[01:08:00:27:3b:eb:45], tid=0x7ba38e78
Sep 17 18:49:43 server.rapavlichenko.net kea-dhcp4[1531]: 2026-09-17 18:49:43.472 INFO [kea-dhcp4.packets/1531.140566014355136] DHCP4_PACKET_RECEIVED [hwty
pe=1 08:00:27:3b:eb:45], cid=[01:08:00:27:3b:eb:45], tid=0x7ba38e78: DHCPREQUEST (type 3) received from 0.0.0.0 to 255.255.255.255 on interface eth1
Sep 17 18:49:43 server.rapavlichenko.net kea-dhcp4[1531]: 2026-09-17 18:49:43.472 INFO [kea-dhcp4.leases/1531.140566014355136] DHCP4_INIT_REBOOT [hwtype=1
08:00:27:3b:eb:45], cid=[01:08:00:27:3b:eb:45], tid=0x7ba38e78: client is in INIT-REBOOT state and requests address 192.168.1.30
Sep 17 18:49:43 server.rapavlichenko.net kea-dhcp4[1531]: 2026-09-17 18:49:43.473 INFO [kea-dhcp4.leases/1531.140566014355136] DHCP4_LEASE_ALLOC [hwtype=1
08:00:27:3b:eb:45], cid=[01:08:00:27:3b:eb:45], tid=0x7ba38e78: lease 192.168.1.30 has been allocated for 7200 seconds
Sep 17 18:49:43 server.rapavlichenko.net kea-dhcp4[1531]: 2026-09-17 18:49:43.473 INFO [kea-dhcp4.leases/1531.140566014355136] DHCP4_LEASE_REUSE [hwtype=1
08:00:27:3b:eb:45], cid=[01:08:00:27:3b:eb:45], tid=0x7ba38e78: lease 192.168.1.30 has been reused for 6001 seconds
Sep 17 18:49:43 server.rapavlichenko.net kea-dhcp4[1531]: 2026-09-17 18:49:43.473 INFO [kea-dhcp4.packets/1531.140566014355136] DHCP4_PACKET_SEND [hwtype=1
08:00:27:3b:eb:45], cid=[01:08:00:27:3b:eb:45], tid=0x7ba38e78: trying to send packet DHCPACK (type 5) from 192.168.1.1:67 to 192.168.1.30:68 on interface
eth1
```

5.5 Проверка интерфейсов

Интерфейс eth1 получил адрес 192.168.1.30/24 из настроенного пула.

```
root@client:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe64:5285 prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:a00:27ff:fe64:5285 prefixlen 64 scopeid 0x0<g
    ether 08:00:27:64:52:85 txqueuelen 1000 (Ethernet)
    RX packets 1850 bytes 158332 (154.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1161 bytes 131509 (128.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.30 netmask 255.255.255.0 broadcast 192.168.1.255
    ether 08:00:27:3b:eb:45 txqueuelen 1000 (Ethernet)
    RX packets 470 bytes 36182 (35.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 688 bytes 136582 (133.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 49 bytes 4957 (4.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 49 bytes 4957 (4.8 KiB)
```

5.6 Файл аренд Кеа

В /var/lib/kea/kea-leases4.csv записана активная аренда адреса 192.168.1.30.

```
root@server:~# cat /var/lib/kea/kea-leases4.csv
address,hwaddr,client_id,valid_lifetime,expire,subnet_id,fqdn_fwd,fqdn_rev,hostname,state,user_context,pool_id
192.168.1.30,08:00:27:3b:eb:45,01:08:00:27:3b:eb:45,4000,1789675380,1,0,0,client,0,,0
root@server:~#
```

6. Настройка DDNS

6.1 Создание TSIG-ключа

Создан ключ DHCP_UPDATER.

```
root@server:~# mkdir -p /etc/named/keys
tsig-keygen -a HMAC-SHA512 DHCP_UPDATER > /etc/named/keys/dhcp_updater.key
root@server:~#
```

Проверено содержимое ключа.

```
root@server:~# cat /etc/named/keys/dhcp_updater.key
key "DHCP_UPDATER" {
    algorithm hmac-sha512;
    secret "k0x9P7PurbWVe3dPHI30WaNgtLAA2/Ps0uztCw+g8XHOHCZHYLfz2b+fu9TF0aHfnSUNGfVXv1hyR1ck9x1X2g==";
};
root@server:~#
```

6.2 Права и подключение ключа

Каталогу назначен владелец
named.

```
root@server:~# chown -R named:named /etc/named/keys  
root@server:~#
```

Ключ подключён в named.conf.

```
type hint;  
file "named.ca";  
};  
  
include "/etc/named.rfc1912.zones";  
include "/etc/named.root.key";  
include "/etc/named/rapavlichenko.net";  
include "/etc/named/keys/dhcp_updater.key";  
|
```

6.3 Разрешение обновления зон

В прямой и обратной зонах разрешены обновления, подписанные ключом DHCP_UPDATER.

```
GNU nano 8.1 /etc/named.conf
zone "rapavlichenko.net" IN {
    type master;
    file "master/fz/rapavlichenko.net";
    update-policy {
        grant DHCP_UPDATER wildcard *.rapavlichenko.net A DHCID;
    };
};

zone "1.168.192.in-addr.arpa" IN {
    type master;
    file "master/rz/192.168.1";
    update-policy {
        grant DHCP_UPDATER wildcard *.1.168.192.in-addr.arpa PTR DHCID;
    };
};
```

6.4 Проверка BIND и подготовка Kea

Конфигурация BIND проверена, служба перезапущена, создан файл /etc/kea/tsig-keys.json.

```
root@server:~# named-checkconf  
root@server:~# systemctl restart named  
root@server:~# touch /etc/kea/tsig-keys.json  
root@server:~#
```

6.5 Перенос TSIG-ключа

Получен секрет созданного ключа.

```
root@server:~# awk -F'"' '{print $2}' /etc/named/keys/dhcp_updater.key  
k0x9P7PurbwVe3dPHi30MaMGtLAA2/Ps0uztCw+g8XKOKCZHYLfz2b+fu9TF0aMfnSUNGfVXv1hyR1ck9x1X2g==  
root@server:~#
```

Ключ сохранён в формате JSON.

```
GNU nano 8.1 /etc/kea/tsig-keys.json  
"tsig-keys": [  
  {  
    "name": "DHCP_UPDATER",  
    "algorithm": "hmac-sha512",  
    "secret": "k0x9P7PurbwVe3dPHi30MaMGtLAA2/Ps0uztCw+g8XKOKCZHYLfz2b+fu9TF0aMfnSUNGfVXv1hyR1ck9x1X2g=="  
  }  
]
```

6.6 Защита файла ключа

Файлу назначен владелец kea, права доступа ограничены значением 640.

```
root@server:~# chown kea:kea /etc/kea/tsig-keys.json
root@server:~# chmod 640 /etc/kea/tsig-keys.json
root@server:~#
```

6.7 Конфигурация Kea DDNS

Заданы параметры обновления прямой и обратной зон через DNS-сервер 192.168.1.1.

```
GNU nano 8.1
{
  "DhcpDdns": {
    "ip-address": "127.0.0.1",
    "port": 53001,
    "control-socket": {
      "socket-type": "unix",
      "socket-name": "/run/kea/kea-ddns-ctrl-socket"
    },

    <? include "/etc/kea/tsig-keys.json" ?>

    "forward-ddns": {
      "ddns-domains": [
        {
          "name": "rapavlichenko.net.",
          "key-name": "DHCP_UPDATER",
          "dns-servers": [
            { "ip-address": "192.168.1.1" }
          ]
        }
      ]
    },

    "reverse-ddns": {
      "ddns-domains": [
        {
```

6.8 Запуск службы DDNS

Конфигурация проверена, служба kea-dhcp-ddns включена и запущена.

```
root@server: # chown kea.kea /etc/kea/kea-dhcp-ddns.conf
root@server: # kea-dhcp-ddns -t /etc/kea/kea-dhcp-ddns.conf
2026-09-17 19:09:06.894 ERROR [kea-dhcp-ddns.dhcpddns/15324.140429421727040] DHCP_DDNS_CONFIG_CHECK_FAIL DHCP-DDNS server configuration check failed: 'socket-name' is invalid: invalid path specified: '/run/kea', supported path is '/var/run/kea'
2026-09-17 19:09:06.902 INFO [kea-dhcp-ddns.dctl/15324.140429421727040] DCTL_CONFIG_CHECK_COMPLETE server has completed configuration check: listening on 127.0.0.1, port 53001, using UDP, result: failure(1), text:'socket-name' is invalid: invalid path specified: '/run/kea', supported path is '/var/run/kea'
Error encountered: 'socket-name' is invalid: invalid path specified: '/run/kea', supported path is '/var/run/kea'
root@server: # systemctl enable --now kea-dhcp-ddns.service
root@server: # systemctl status kea-dhcp-ddns.service --no-pager
● kea-dhcp-ddns.service - Kea DHCP-DDNS Server
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp-ddns.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-09-17 17:39:12 UTC; 1h 30min ago
  Invocation: 4b2a578d42d74aaa8b9d13fbec61051
     Docs: man:kea-dhcp-ddns(8)
    Main PID: 15330 (kea-dhcp-ddns)
      Status: "Dispatching packets..."
        Tasks: 5 (limit: 10260)
       Memory: 6.7M (peak: 6.9M)
          CPU: 357ms
      CGroup: /system.slice/kea-dhcp-ddns.service
              └─15330 /usr/sbin/kea-dhcp-ddns -c /etc/kea/kea-dhcp-ddns.conf

Sep 17 18:29:44 server.rapavlichenko.net kea-dhcp-ddns[15330]: 2026-09-17 18:29:44.782 ERROR [kea-dhcp-ddns.d2-to-dns/15330.140598997429568] DHCP_DDNS_FORMAR...
Sep 17 18:29:44 server.rapavlichenko.net kea-dhcp-ddns[15330]: 2026-09-17 18:29:44.791 ERROR [kea-dhcp-ddns.d2-to-dns/15330.140598997429568] DHCP_DDNS_ADD_FA...
Sep 17 18:29:44 server.rapavlichenko.net kea-dhcp-ddns[15330]: Forward Change: yes
Sep 17 18:29:44 server.rapavlichenko.net kea-dhcp-ddns[15330]: Reverse Change: yes
Sep 17 18:29:44 server.rapavlichenko.net kea-dhcp-ddns[15330]: FQDN: [client.rapavlichenko.net.]
Sep 17 18:29:44 server.rapavlichenko.net kea-dhcp-ddns[15330]: IP Address: [192.168.1.38]
Sep 17 18:29:44 server.rapavlichenko.net kea-dhcp-ddns[15330]: DHCID: [000101914194EC0505483008512A18187741FCBA701BA5EE80150CE0BA4C91CA116CA]
Sep 17 18:29:44 server.rapavlichenko.net kea-dhcp-ddns[15330]: Lease Expires On: 20260917190944
Sep 17 18:29:44 server.rapavlichenko.net kea-dhcp-ddns[15330]: Lease Length: 2400
Sep 17 18:29:44 server.rapavlichenko.net kea-dhcp-ddns[15330]: Conflict Resolution Mode: check-with-dhcid
Hint: Some lines were ellipsized, use -l to show in full.
root@server: #
```

6.9 Включение DDNS в DHCP

В `kea-dhcp4.conf` включены обновления и указан домен `rapavlichenko.net`.

```
{
  "name": "domain-name-servers",
  "data": "192.168.1.1"
},
"dhcp-ddns": {
  "enable-updates": true
},
"ddns-qualifying-suffix": "rapavlichenko.net",
"ddns-override-client-update": true,
"subnet4": [
  {
    "id": 1,
    "subnet": "192.168.1.0/24",
    "pools": [
      {
        "pool": "192.168.1.30 - 192.168.1.199"
      }
    ]
  }
],
```

6.10 Перезапуск ДНСР-сервера

Журнал подтвердил успешную работу Кеа после включения динамических обновлений.

```
root@server:~# kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
2026-09-17 19:12:06.720 WARN [kea-dhcp4.dhcpsrv/15494.139852469934208] DHCP4_SRV_MT_DISABLED_QUEUE_CONTROL disabling dhcp queue control when multi-threading is enabled.
2026-09-17 19:12:06.729 WARN [kea-dhcp4.dhcp4/15494.139852469934208] DHCP4_RESERVATIONS_LOOKUP_FIRST_ENABLED Multi-threading is enabled and host reservations lookup is always performed first.
2026-09-17 19:12:06.730 INFO [kea-dhcp4.dhcpsrv/15494.139852469934208] DHCP4_SRV_CFGMGR_NEW_SUBNET4 a new subnet has been added to configuration: 192.168.0.0/24 with params: t1=1000, t2=2000, valid-lifetime=4000
2026-09-17 19:12:06.730 INFO [kea-dhcp4.dhcpsrv/15494.139852469934208] DHCP4_SRV_CFGMGR_SOCKET_TYPE_SELECT using socket type raw
2026-09-17 19:12:06.737 INFO [kea-dhcp4.dhcpsrv/15494.139852469934208] DHCP4_SRV_CFGMGR_ADD_IFACE listening on interface eth1
2026-09-17 19:12:06.737 INFO [kea-dhcp4.dhcpsrv/15494.139852469934208] DHCP4_SRV_CFGMGR_SOCKET_TYPE_DEFAULT "dhcp-socket-type" not specified, using default socket type raw
2026-09-17 19:12:06.737 INFO [kea-dhcp4.dhcpsrv/15494.139852469934208] DHCP4_SRV_LEASE_MGR_BACKENDS_REGISTERED the following lease backend types are available: memfile
2026-09-17 19:12:06.737 INFO [kea-dhcp4.hosts/15494.139852469934208] HOSTS_BACKENDS_REGISTERED the following host backend types are available:
2026-09-17 19:12:06.737 INFO [kea-dhcp4.dhcpsrv/15494.139852469934208] DHCP4_SRV_FORENSIC_BACKENDS_REGISTERED the following forensic backend types are available:
2026-09-17 19:12:06.737 INFO [kea-dhcp4.database/15494.139852469934208] CONFIG_BACKENDS_REGISTERED the following config backend types are available:
root@server:~# systemctl restart kea-dhcp4.service
root@server:~# systemctl status kea-dhcp4.service
● kea-dhcp4.service - Kea DHCPv4 Server
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-09-17 19:12:18 UTC; 4s ago
   Invocation: dc15fa6b8a4a223eafc235f995232c7a
     Docs: man:kea-dhcp4(8)
    Main PID: 15504 (kea-dhcp4)
   Status: "Dispatching packets..."
     Tasks: 6 (limit: 10260)
    Memory: 2.4M (peak: 6.8M)
       CPU: 149ms
    CGroup: /system.slice/kea-dhcp4.service
            └─15504 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf

Sep 17 19:12:18 server.rapavlichenko.net kea-dhcp4[15504]: 2026-09-17 19:12:18.888 INFO [kea-dhcp4.dhcpsrv/15504.140175716448384] DHCP4_SRV_MEMFILE_DB opened
Sep 17 19:12:18 server.rapavlichenko.net kea-dhcp4[15504]: 2026-09-17 19:12:18.889 INFO [kea-dhcp4.dhcpsrv/15504.140175716448384] DHCP4_SRV_MEMFILE_LEASES loaded
Sep 17 19:12:18 server.rapavlichenko.net kea-dhcp4[15504]: 2026-09-17 19:12:18.908 INFO [kea-dhcp4.dhcpsrv/15504.140175716448384] DHCP4_SRV_MEMFILE_LEASES loaded
Sep 17 19:12:18 server.rapavlichenko.net kea-dhcp4[15504]: 2026-09-17 19:12:18.909 INFO [kea-dhcp4.dhcpsrv/15504.140175716448384] DHCP4_SRV_MEMFILE_EXTRACTION started
Sep 17 19:12:18 server.rapavlichenko.net kea-dhcp4[15504]: 2026-09-17 19:12:18.910 INFO [kea-dhcp4.dhcpsrv/15504.140175716448384] DHCP4_SRV_MEMFILE_LFC_SENDING started
Sep 17 19:12:18 server.rapavlichenko.net kea-dhcp4[15504]: 2026-09-17 19:12:18.913 INFO [kea-dhcp4.dhcpsrv/15504.140175716448384] DHCP4_SRV_DNS_SENDING started
Sep 17 19:12:18 server.rapavlichenko.net kea-dhcp4[15504]: 2026-09-17 19:12:18.927 INFO [kea-dhcp4.dhcpsrv/15504.140175716448384] DHCP4_SRV_CFGMGR_USE_ALL started
Sep 17 19:12:18 server.rapavlichenko.net kea-dhcp4[15504]: 2026-09-17 19:12:18.929 INFO [kea-dhcp4.dhcp4/15504.140175716448384] DHCP4_MULTI_THREADING_INITIALIZED
Sep 17 19:12:18 server.rapavlichenko.net kea-dhcp4[15504]: 2026-09-17 19:12:18.930 INFO [kea-dhcp4.dhcp4/15504.140175716448384] DHCP4_STARTED Kea DHCPv4 Server
Sep 17 19:12:18 server.rapavlichenko.net systemd[1]: Started kea-dhcp4.service - Kea DHCPv4 Server.
```

7. Результаты

7.1 Повторное получение адреса

Клиентское соединение eth1 переподключено для обновления аренды и DNS-записи.

```
root@client:~# nmcli connection down eth1
Connection 'eth1' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/9)
root@client:~# nmcli connection up eth1
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/10)
root@client:~#
```

7.2 Журнал динамической зоны

Появление файла .jnl подтвердило принятие динамического обновления сервером BIND.

```
root@server:~# dig @192.168.1.1 client.rapavlichenko.net
ls -l /var/named/master/fz/rapavlichenko.net.jnl

; <<>> DiG 9.18.33 <<>> @192.168.1.1 client.rapavlichenko.net
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 41928
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 1232
;; COOKIE: 3a77260f54569814010000006aac3ed34f404fa4a8b6b443 (good)
;; QUESTION SECTION:
;client.rapavlichenko.net.      IN      A

;; ANSWER SECTION:
client.rapavlichenko.net. 1333 IN      A      192.168.1.30

;; Query time: 0 msec
;; SERVER: 192.168.1.1#53(192.168.1.1) (UDP)
;; WHEN: Thu Sep 17 19:26:11 UTC 2026
;; MSG SIZE rcvd: 97

-rw-r--r--. 1 named named 843 Sep 17 19:26 /var/named/master/fz/rapavlichenko.net.jnl
root@server:~#
```

7.3 Проверка DNS-записи клиента

Команда `dig` вернула A-запись `client.rapavlichenko.net` с адресом `192.168.1.30`.

```
root@client:~# dig @192.168.1.1 client.rapavlichenko.net

; <<>> DiG 9.18.33 <<>> @192.168.1.1 client.rapavlichenko.net
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 20334
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL:
;

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 553aa8cb13d74a78010000006aac3f2963bbb2597b934a07 (good)
;; QUESTION SECTION:
;client.rapavlichenko.net.      IN      A

;; ANSWER SECTION:
client.rapavlichenko.net. 1333 IN      A      192.168.1.30

;; Query time: 9 msec
;; SERVER: 192.168.1.1#53(192.168.1.1) (UDP)
;; WHEN: Thu Sep 17 19:27:37 UTC 2026
;; MSG SIZE rcvd: 97

root@client:~#
```

8. Автоматизация

8.1 Сохранение конфигурации DHCP

Рабочие файлы Кеа скопированы в каталог `provision/server/dhcp`.

```
root@server:~# cd /vagrant/provision/server
root@server:/vagrant/provision/server# mkdir -p /vagrant/provision/server/dhcp/etc/kea
root@server:/vagrant/provision/server# cp -R /etc/kea/* /vagrant/provision/server/dhcp/etc/kea/
root@server:/vagrant/provision/server#
```

8.2 Обновление конфигурации DNS

Фактические файлы BIND скопированы в каталог автоматического развёртывания.

```
root@server:/vagrant/provision/server# cp -R /etc/kea/* /vagrant/provision/server/dhcp/etc/kea/
root@server:/vagrant/provision/server# cd /vagrant/provision/server/dns/
cp -R /var/named/* /vagrant/provision/server/dns/var/named/
cp -R /etc/named/* /vagrant/provision/server/dns/etc/named/
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/rapavlichenko.net'?
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'?
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.ca': Protocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.empty': Protocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.localhost': Protocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.loopback': Protocol error
cp: overwrite '/vagrant/provision/server/dns/etc/named/rapavlichenko.net'?
root@server:/vagrant/provision/server/dns#
```

8.3 Создание сценария dhcp.sh

В каталоге `provision/server` создан исполняемый сценарий настройки DHCP.

```
root@server:/vagrant/provision/server/dns# cd /vagrant/provision/server
root@server:/vagrant/provision/server# touch dhcp.sh
root@server:/vagrant/provision/server# chmod +x dhcp.sh
root@server:/vagrant/provision/server# nano dhcp.sh
```

8.4 Содержимое сценария

Сценарий устанавливает Kea, копирует конфигурацию, настраивает права, SELinux и firewalld, затем запускает службы DHCP и DDNS.

```
GNU nano 8.1
#!/bin/bash
echo "Provisioning script $0"

echo "Install needed packages"
dnf -y install kea

echo "Copy configuration files"
cp -R /vagrant/provision/server/dhcp/etc/kea/* /etc/kea/

echo "Fix permissions"
chown -R kea:kea /etc/kea
chmod 640 /etc/kea/tsig-keys.json
restorecon -vR /etc
restorecon -vR /var/lib/kea

echo "Configure firewall"
firewall-cmd --add-service=dhcp
firewall-cmd --add-service=dhcp --permanent

echo "Start dhcpd service"
systemctl --system daemon-reload
systemctl enable --now kea-dhcp4.service
systemctl enable --now kea-dhcp-ddns.service
```

8.5 Подключение сценария к серверу

Provisioner server dhcp запускает сценарий после настройки DNS.

```
server.vm.provision "server dns",  
  type: "shell",  
  preserve_order: true,  
  path: "provision/server/dns.sh"  
  
server.vm.provision "server dhcp",  
  type: "shell",  
  preserve_order: true,  
  path: "provision/server/dhcp.sh"  
  
end  
  
config.vm.define "client", autostart: false do |  
  client.vm.box = "rockylinux10"
```

8.6 Завершение работы

После проверки клиентская и серверная виртуальные машины выключены.

```
C:\work\rapavlichenko>vagrant halt client  
==> client: Attempting graceful shutdown of VM...
```

9. Контрольные вопросы

9.1 DHCP и сетевые подключения

- NetworkManager хранит профили в `/etc/NetworkManager/system-connections/`.
- DHCP автоматически выдаёт IP-адрес, маску, шлюз, DNS-серверы и срок аренды.
- Основной обмен выполняется сообщениями DHCPDISCOVER, DHCPOFFER, DHCPREQUEST и DHCPACK.
- Сервер использует UDP-порт 67, клиент — порт 68.

9.2 Файлы Kea и DDNS

- `/etc/kea/kea-dhcp4.conf` содержит настройки DHCPv4.
- `/etc/kea/kea-dhcp-ddns.conf` задаёт взаимодействие с DNS.
- `/var/lib/kea/kea-leases4.csv` хранит сведения об арендах.
- DDNS автоматически создаёт и обновляет A- и PTR-записи.
- TSIG-ключ защищает запросы динамического обновления.

9.3 Диагностические утилиты

- `ifconfig` показывает адреса, состояние и статистику интерфейсов.
- `ifconfig -a` выводит также отключённые интерфейсы.
- `ping -c 4 192.168.1.1` проверяет доступность четырьмя запросами.
- Итоговая статистика `ping` показывает задержку и потери пакетов.

10. Выводы

10.1 Выводы

В ходе работы были получены практические навыки установки и конфигурирования DHCP-сервера Kea. Сервер автоматически выдал клиенту сетевые параметры, а настройка DDNS обеспечила создание записи `client.rapavlichenko.net`. Сценарии Vagrant сохранили выполненную конфигурацию для повторного развёртывания стенда.

11. Завершение

Спасибо за внимание